

Load Interconnection Request System Impact Study Report

For Queue Project Q22-2017

Rogers Lake 230 kV
May 21, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2017. The Interconnection Customer proposes a 670.5 MW large load (data center) facility interconnecting at the Rogers Lake 230 kV substation in the Ohio area. The request was received on February 24, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 638 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$41,506,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rogers Lake 138 kV/1 kV Auto	Gilstrap - Rogers Lake 230.0 kV	248.0	200	100.0
Rogers Lake 230 kV/1 kV Autotransformer	Gilstrap - Rogers Lake 230.0 kV	193.8	256	100.0
Gilstrap 230/138 kV Transformer	Gilstrap - Rogers Lake Substation 230.0 kV	136.0	187	23.0
Nelson - Lark 100.0 kV	Gilstrap - Rogers Lake 230.0 kV Line	120.9	115	9.1
Rogers Lake - Gilstrap 230.0 kV	Rogers Lake 230/1 kV Autotransformer	107.3	463	100.0
Gilstrap Substation - Stoner 230.0 kV L	El Paso - Ravenna 500.0 kV	104.1	483	24.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rogers Lake 138 kV/1 kV Auto	Replace transformer with higher-rated unit	\$5,870,000	78% (\$4,593,000)
Rogers Lake 230 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$13,577,000	78% (\$10,624,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gilstrap 230/138 kV TX	Replace transformer with higher-rated unit	\$7,391,000	68% (\$5,012,000)
Nelson - Lark 100.0 kV	Rebuild line with higher-capacity conductor	\$4,763,000	60% (\$2,861,000)
Rogers Lake - Gilstrap 230.0 kV	Reconductor line	\$9,713,000	78% (\$7,600,000)
Gilstrap - Stoner 230.0 kV	Reconductor line	\$16,218,000	67% (\$10,816,000)
	Total allocated to project		\$41,506,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.